Metal / plastic classifier — training record

ResNeXt-50 (32×4d), two classes · Checkpoint `models/resnext50_metal_plastic_belt_stage1_layer4.pt`
Trained 5 October 2026 on a Colab T4 · Record prepared 16 October 2026

Dataset composition

10,806 images, stored at 224×224. Train, validation, and test share neither a filename nor file bytes. The test split is balanced on purpose: 150 metal and 150 plastic.

Split	Stage 1 metal	Belt metal	Stage 1 plastic	Belt plastic	Total
Train	680	680	3,949	3,946	9,255
Validation	40	93	406	712	1,251
Test	40	110	60	90	300
Total	**760**	**883**	**4,415**	**4,748**	**10,806**

Plastic is PET, HDPE, and PP. Metal is the non-plastic bucket: belt cans, plus stage 1 photos labeled OTHERS (foil trays and other rejects). A file in that folder is not a claim that the object is metallic. Belt frames run from 11 September through 30 September 2026. Cans from 28–30 September are held out in contiguous time windows, with a two-frame gap left out on each side so a near-duplicate does not sit in two splits. Training is about 5.8 plastic images for each metal image.

Training procedure

The network is ResNeXt-50 (32×4d). The 1,000-way ImageNet head is replaced by a linear layer from 2,048 features to two classes. This checkpoint was fine-tuned in two stages on the set above, each stage resuming the weights of the one before it. The run started from the pinned Drive head-only checkpoint (validation 99.07%, test 98.87% on the earlier Drive set), not from a fresh ImageNet initialization.

Stage 1 froze every residual stage and trained only the classification head, for 6 epochs at learning rate 0.0003. The best epoch reached 95.61% balanced validation accuracy (metal 123/133, plastic 1,104/1,118) and 96.67% balanced test accuracy (142/150 metal, 148/150 plastic).

Stage 2 unfroze the last residual stage, layer4, and left layer1 through layer3 frozen. It ran 4 epochs, with the head still at 0.0003 and layer4 at 0.00003. The file on disk is the epoch with the highest balanced validation accuracy, the mean of metal recall and plastic recall. That was epoch 3, at 99.11% (metal 131/133, plastic 1,115/1,118). Epoch 4 matched those counts and was not kept: the saver replaces the checkpoint only when the balanced score rises.

Training parameters

Optimizer	AdamW, weight decay 0.01 (PyTorch default). No learning-rate schedule.
Loss	Cross-entropy. Training weights are inverse class frequency raised to 0.5: metal 1.845, plastic 0.766. Validation and test loss are unweighted.
Batch size	32
Input	224×224, then ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).
Augmentation	Training only: random resized crop (scale 0.6–1.0, aspect 0.9–1.1), horizontal flip, rotation ±15°. Evaluation uses the full frame, bilinear-resized to 224×224.
Selection	Highest balanced validation accuracy.
Device	NVIDIA T4 (Google Colab).

Test results

The holdout is 300 images and was scored once, after validation had already chosen the checkpoint. It was not used to pick the epoch.

	Metal	Plastic	Overall
Images	150	150	300
Correct	149	150	299
Recall	99.33%	100%	—
Precision	100%	99.34%	—
F1	99.67%	99.67%	99.67% macro
Accuracy	—	—	99.67%
Balanced accuracy	—	—	99.67%

Confusion matrix. Rows are the true class. Columns are the prediction. Counts are the 300 test images.

	Predicted metal	Predicted plastic
True metal	**149**	1
True plastic	0	**150**

The single error is a crushed can from 29 September 2026, `capture_01896_20260929_141207_925981.jpg`, called plastic at 69%. A frame from the same burst, four seconds earlier, was classified correctly.

Validation, used only to choose the epoch, was 1,246 of 1,251 images (99.60% accuracy, 99.11% balanced). Five images were wrong: three plastic called metal (two HDPE, one PP) and two metal called plastic (both OTHERS).

The file `confusion_matrix.png` at the repository root is an earlier Drive run (596 images, test accuracy 98.11%). It is not this checkpoint.